

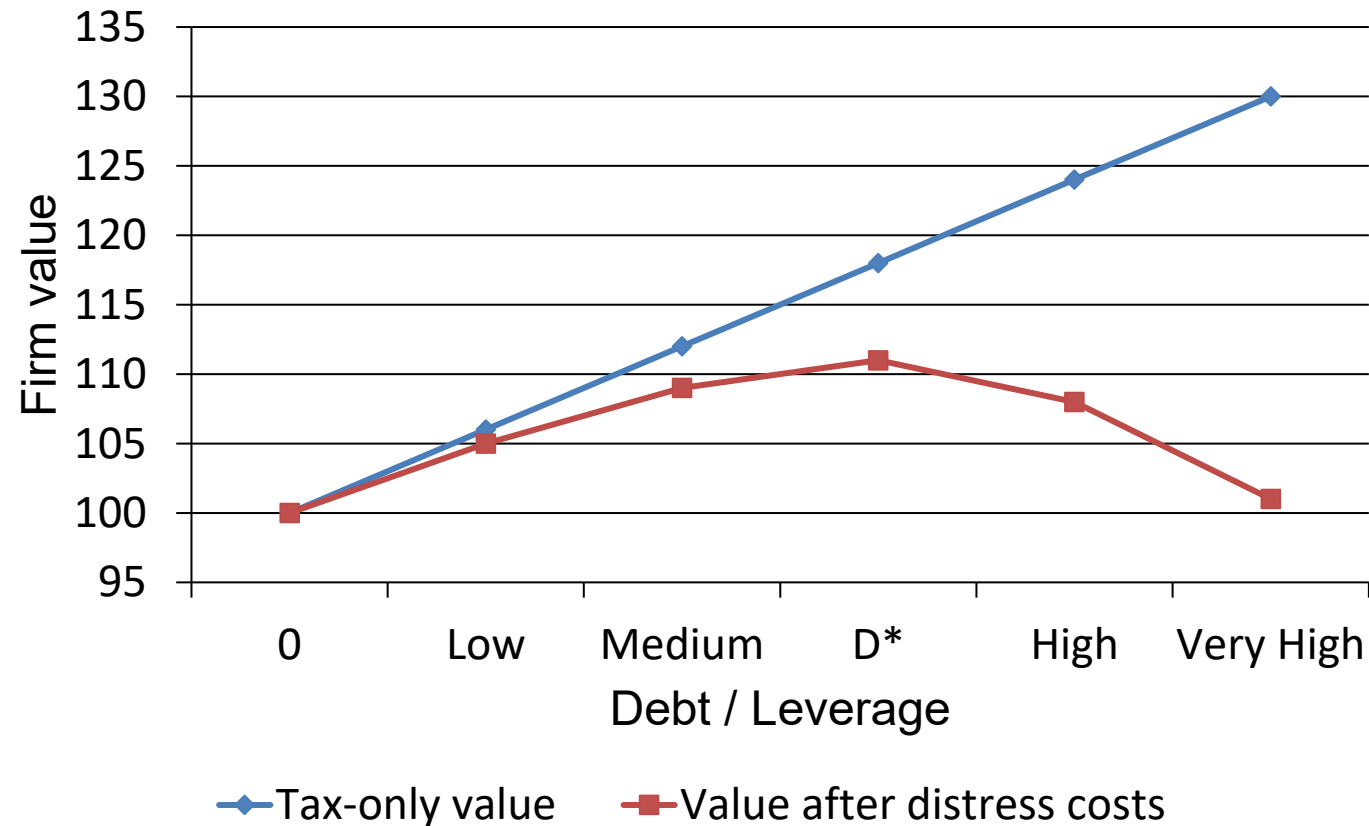
Capital Structure Policy

Capital Structure and Dividend Policy



Recap — The MM Case III Framework

Starting point: tax benefits versus expected financial distress costs.



CASE II

Tax shield pushes firm value upward.

CASE III

Expected distress costs eventually offset the tax benefit.

D*

Firm value is maximized at the optimal leverage level.

Today's question: what determines the shape and size of those distress and incentive costs?

Leverage = Amplification

Fixed costs magnify the effect of changes in activity on profits.

OPERATING LEVERAGE

Fixed operating costs amplify changes in sales into larger changes in EBIT.

Examples: factories, aircraft fleets, stores, data centres.

FINANCIAL LEVERAGE

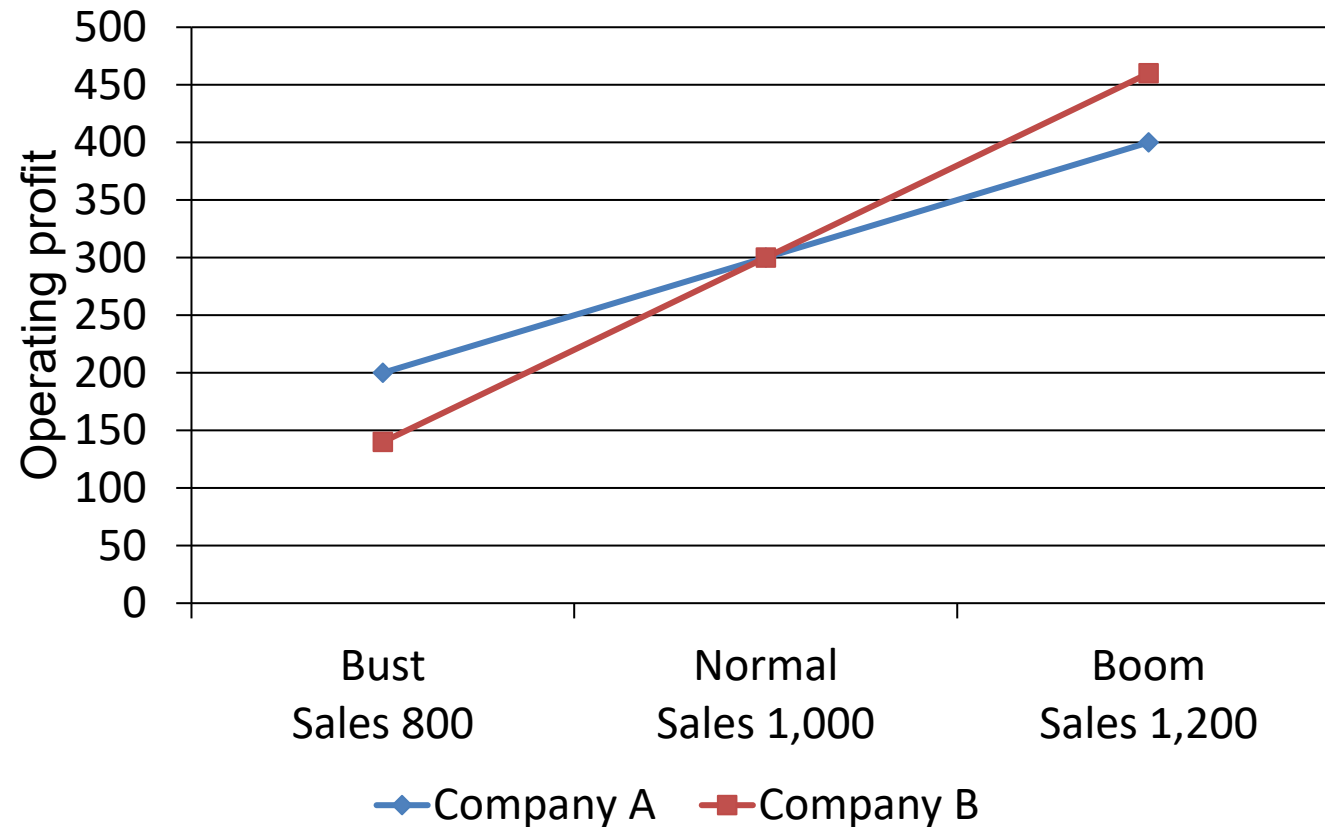
Fixed financing costs — especially interest — amplify changes in EBIT into larger changes in earnings available to equity.

More debt → greater financial risk.

The same mechanism creates upside in good states and sharper downside in bad states.

Same Sales Shock — Different Profit Effect

Two companies start with the same sales and profit, but different cost structures.



COMPANY A

Variable cost 50%
Fixed cost 200

COMPANY B

Variable cost 20%
Fixed cost 500

KEY POINT

Same normal profit (€300), but B has greater operating leverage.

±20% sales → Company A profit: ±33% | Company B profit: approximately ±53%

Now Add Debt — What Happens to Shareholders?

Stay with Company B: interest is fixed, so shareholders receive the residual.

	BUST	NORMAL	BOOM
EBIT	€140	€300	€460
Interest	€120	€120	€120
Income left for shareholders	€20	€180	€340

EBIT

€140 ← €300 → €460

Equity income

€20 ← €180 → €340

KEY IDEA

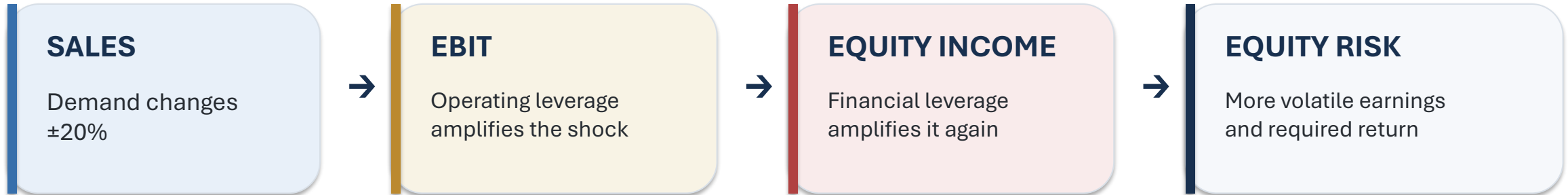
Interest stays fixed.
Shareholders absorb the variation.

What happens if EBIT falls below €120?

That question takes us directly to financial distress.

Business Risk + Financial Leverage = Greater Equity Risk

Financial risk is added on top of the firm's existing business risk.



High operating leverage + high debt can make a firm especially vulnerable to a downturn.

CAPITAL-STRUCTURE IMPLICATION

Higher business risk → lower capacity to safely take financial risk.

Opening the Black Box — What Creates Financial Distress Costs?

MM Case III used expected distress costs; now we examine what sits inside that term.

DIRECT COSTS

- Legal fees
- Restructuring advisers
- Court / administration
- Valuation and accounting costs

INDIRECT COSTS

- Lost customers
- Tighter supplier terms
- Employee departures
- Fire-sale asset discounts

LOST FLEXIBILITY

- Covenant constraints
- Reduced investment capacity
- Harder refinancing
- Management distraction

MM Case III tells us distress costs matter; now we ask why they differ across firms.

Which Businesses Are Most Sensitive to Distress?

Indirect costs depend on what the firm needs to preserve in order to operate.

CUSTOMER TRUST

Warranties
Prepayments
Long-term service
Brand confidence

HUMAN CAPITAL

Consultants
Engineers
Sales teams
Key managers

SUPPLY CHAIN

Trade credit
Critical suppliers
Inventory continuity

ASSET RECOVERY

Specialized assets
Intangibles
Fire-sale discounts

Who can safely carry more debt — a regulated utility or an airline? What evidence do you need?

Expected Distress Cost — The Number That Matters for Capital Structure

A large bankruptcy loss matters only to the extent that distress is sufficiently likely.

CORE IDEA

Expected Distress Cost = Probability of Distress × Economic Cost if Distress Occurs

EXAMPLE A

1% probability × €100m loss
= €1m expected cost

EXAMPLE B

10% probability × €100m loss
= €10m expected cost

LEVERAGE EFFECT

More debt can increase both default probability and the damage created by distress.

This is the bridge from the MM Case III graph to firm-specific debt capacity.

What Pushes Expected Distress Costs Up or Down?

Use these determinants when analysing a real company.

CASH-FLOW RISK

- ↑ Cyclicalit
- ↑ Operating leverage
- ↑ Commodity exposure

More volatility → higher distress probability

ASSET TYPE

- Tangible collateral
- Resale value
- Asset specificity

Recoverable assets → lower loss given distress

STAKEHOLDERS

- Customer confidence
- Supplier credit
- Employee mobility

Sensitive stakeholders → larger indirect costs

LIQUIDITY

- Cash buffer
- Debt maturities
- Refinancing access

More liquidity → more time to adjust

GROWTH OPTIONS

- R&D
- Future projects
- Intangibles

More options → more value can be lost

Use evidence to argue whether your firm's expected distress cost is relatively high or low.

From MM Case III Intuition to Quantification

Now put numbers on the tax benefit and expected distress cost.

MM Case III gave us the graph. The Make-or-Break example shows exactly where the value gain and value loss come from.

STEP 1

Value the operating business without debt.

STEP 2

Add the present value of the expected tax shield.

STEP 3

Subtract the present value of expected distress costs.

TRADE-OFF

$$VL = VU + PV(\text{Tax Shield}) - PV(\text{Expected Distress Costs})$$

Example — Two Firms, Same Debt Ratio

Which firm should be more conservative with debt?

FIRM A — CLOUD SOFTWARE

- Recurring subscriptions
- High gross margins
- Few tangible assets
- Customer switching possible
- Value depends heavily on employees and technology

FIRM B — ELECTRIC UTILITY

- Stable regulated demand
- Tangible infrastructure
- Predictable cash flows
- Assets retain collateral value
- Lower operating volatility

Which firm is likely to have the higher expected distress cost at the same leverage?

Worked Example — Make-or-Break Co.

A simple numerical example shows how tax shields and distress costs compete.

OPERATING CASH FLOW

Boom: \$2,000

Bust: \$100

Probability of each state: 50%

Tax rate: 35%

Risk-free rate: 2%

Required return on operating assets:
10%

DEBT

Debt outstanding: \$1,200

Interest rate: 10%

Promised interest: \$120

In the bust state, only \$100 is
available.

DISTRESS COSTS

If distress occurs:

Direct cost = \$50

Indirect cost = \$50

Total distress cost in the bust state =
\$100

We will value the firm under three financing scenarios.

Make-or-Break — Step 1: Value Without Debt

First establish the value of the operating business.

EXPECTED AFTER-TAX CASH FLOW

$$[0.5 \times \$2,000 + 0.5 \times \$100] \times (1 - 0.35) = \$682.5$$

UNLEVERED VALUE

$$VU = \$682.5 / 0.10 = \$6,825$$

INTERPRETATION

\$6,825 is our benchmark operating value before any financing benefits or distress costs.

Make-or-Break — Step 2: Add Debt, Ignore Distress Costs

Debt creates value through the expected interest tax shield.

EXPECTED TAX SHIELD

$$35\% \times [0.5 \times \$120 + 0.5 \times \$100] = \$38.5 \text{ per year}$$

PV OF TAX SHIELD

$$\$38.5 / 10\% = \$385$$

LEVERED VALUE — NO DISTRESS COST

$$VL = \$6,825 + \$385 = \$7,210$$

So far, leverage adds \$385 of value.

Make-or-Break — Step 3: Add Financial Distress Costs

The same debt now destroys value in the bust state.

PV OF DISTRESS COST

$$\text{Expected annual distress cost} = 0.5 \times \$100 = \$50 \rightarrow \text{PV} = \$50 / 10\% = \$500$$

VALUE WITH DEBT AND DISTRESS COSTS

$$VL = \$6,825 + \$385 - \$500 = \$6,710$$

TAX BENEFIT

+ \$385

DISTRESS COST

– \$500

NET EFFECT

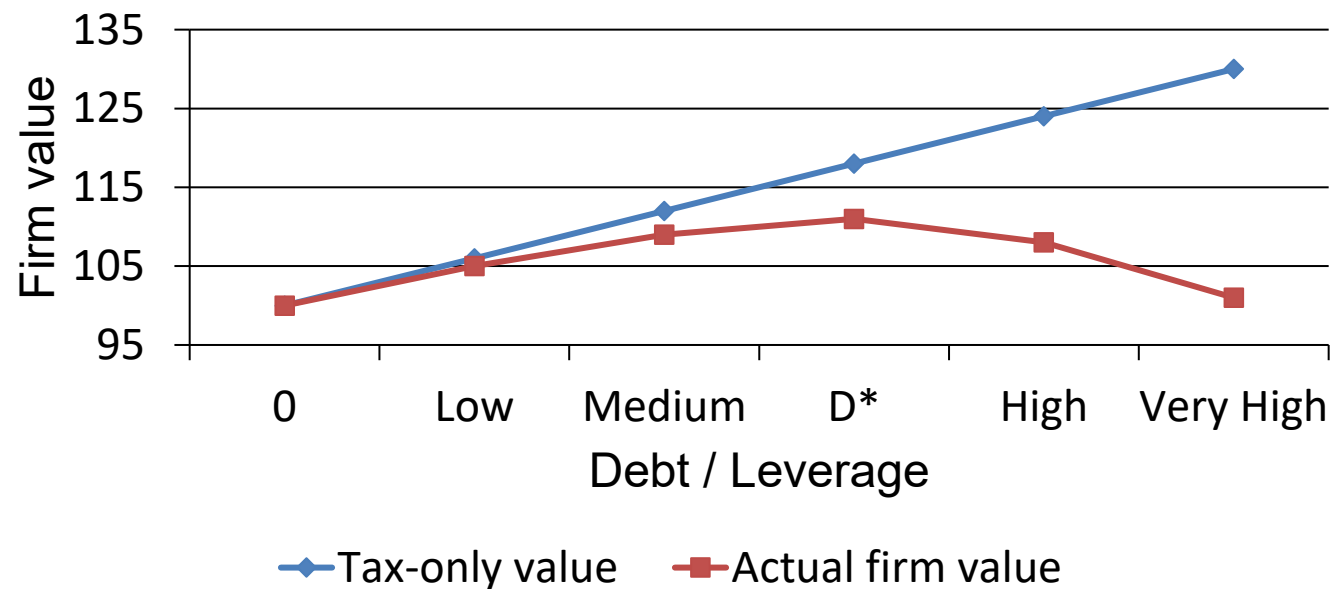
– \$115 vs. unlevered

Trade-Off Theory — The Core Valuation Equation

Optimal leverage balances the marginal benefit and marginal cost of debt.

FOUNDATION

$$VL = VU + PV(\text{Tax Shield}) - PV(\text{Financial Distress Costs})$$



BEFORE D*

Marginal tax benefit > marginal distress cost.

AT D*

Firm value is maximized.

AFTER D*

Marginal distress cost dominates.

What Determines the Optimal Debt Capacity?

The trade-off is different for different firms.

OPERATING RISK

Stable cash flows → more debt capacity

Volatile cash flows → less debt capacity

ASSET TYPE

Tangible, saleable assets → lower distress losses

Intangibles → higher losses

TAX POSITION

More taxable income → more usable tax shields

Low EBIT → limited benefit

STAKEHOLDERS

Customer, supplier and employee sensitivity can amplify indirect costs.

These factors explain why your firm's debt capacity should be high, medium or low.

Group Case — AeroNova: Debt-Funded Share Repurchase?

Work in groups for 10 minutes. Be ready to defend your recommendation — there is no single mechanical answer.

THE COMPANY

European aircraft-components supplier

Revenue: €3.2bn

EBIT: €320m

Current debt: €900m

Cash: €250m

Interest expense: €45m

Tax rate: 25%

60% of sales come from 4 major customers.

THE PROPOSAL

Management proposes €1.0bn of new debt to repurchase shares.

Argument:

- interest is tax deductible
- current coverage is strong
- shares appear undervalued

After the transaction, debt would rise to €1.9bn.

NEW INFORMATION

A major aircraft program may be delayed.

30% of assets are specialized machinery.

Customers require reliable long-term delivery.

€600m of debt matures within 3 years after the proposed transaction.

Decision: Approve €1.0bn / Approve a smaller amount / Reject — give 3 reasons.

AeroNova — Questions to Structure Your Recommendation

Use the framework, not intuition alone.

1 BENEFIT

How does additional debt create value?

Is AeroNova likely to use the tax shield?

2 DISTRESS

What increases the probability of distress?

What could make distress especially costly?

3 FLEXIBILITY

Does the transaction leave enough capacity for a downturn, refinancing or investment?

4 EVIDENCE

What additional information would you request before approving the transaction?

Your recommendation should explicitly connect the financing choice to firm value.

Agency Problems — Why Stakeholders May Want Different Decisions

Debt changes incentives, not only cash flows.

SHAREHOLDERS

Limited liability means their downside is capped, while they keep much of the upside.

CREDITORS

Receive fixed promised payments and therefore care strongly about downside risk.

MANAGERS

May have private benefits, career concerns or incentives that differ from total firm value.

Agency problem = a decision that benefits one stakeholder group but can reduce total firm value.

Agency Cost 1 — Risk Shifting / Asset Substitution

When leverage is high, shareholders may prefer a riskier project even if it has lower total value.

PROJECT 1 — SAFER

50%: \$1,200

50%: \$500

Expected total payoff = \$850

Debt due next year = \$500

PROJECT 2 — RISKIER

25%: \$2,100

75%: \$100

Expected total payoff = \$600

Debt due next year = \$500

Before calculating stakeholder payoffs: which project maximizes total firm value?

Risk Shifting — Who Gets the Payoff?

Limited liability changes shareholders' preferences.

PROJECT 1 — SAFER

Debt payoff:

Success \$500 | Failure \$500 | Expected \$500

Equity payoff:

Success \$700 | Failure \$0 | Expected \$350

Total expected payoff = \$850

PROJECT 2 — RISKIER

Debt payoff:

Success \$500 | Failure \$100 | Expected \$200

Equity payoff:

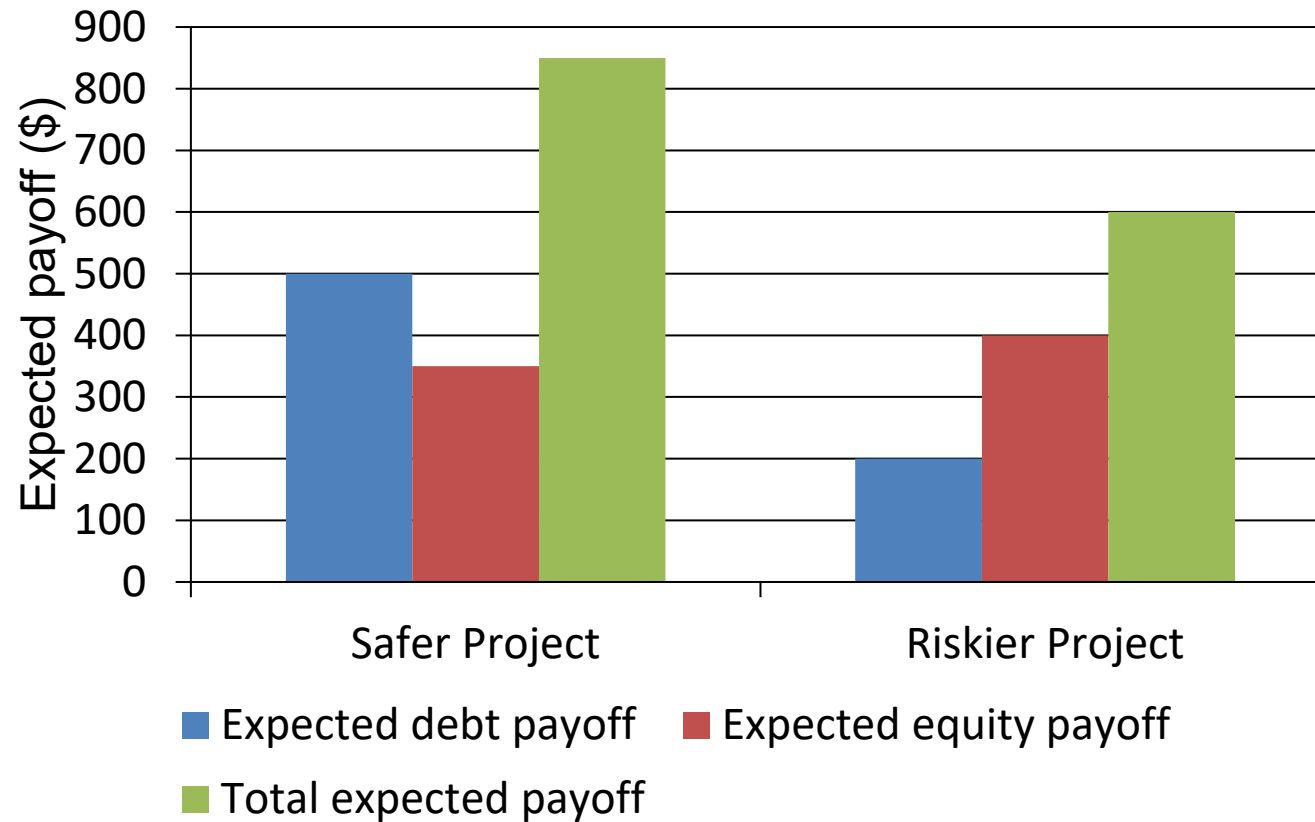
Success \$1,600 | Failure \$0 | Expected \$400

Total expected payoff = \$600

Shareholders prefer Project 2 (\$400 > \$350), even though it destroys \$250 of total expected value.

Risk Shifting — Visual Intuition

As leverage rises, shareholders can gain from volatility at creditors' expense.



TOTAL VALUE

Falls from \$850 to \$600.

EQUITY VALUE

Rises from \$350 to \$400.

WHO LOSES?

Creditors absorb the downside.

Agency Cost 2 — Debt Overhang / Underinvestment

High existing debt can make shareholders reject a positive-NPV project.

EXISTING POSITION

Debt due next year: \$100,000

Existing assets next year: \$50,000

Equity is currently out of the money.

NEW PROJECT

Investment today: \$10,000

Future cash flow:

50%: \$50,000

50%: \$30,000

Expected payoff = \$40,000

ECONOMIC VALUE

The project costs \$10,000 and creates expected cash flow of \$40,000.

It is strongly positive NPV in the simplified example.

Why might shareholders refuse to provide the \$10,000 needed to fund this positive-NPV project?

Debt Overhang — Where Does the New Value Go?

The project improves creditor recovery but leaves equity with no payoff in this example.

WITHOUT PROJECT

Firm value next year: \$50,000

Debt receives: \$50,000

Equity receives: \$0

WITH PROJECT

Firm value next year:

Success \$100,000 | Failure \$80,000

Expected \$90,000

Debt receives all of it.

Equity still receives \$0.

New shareholders pay \$10,000 today but receive \$0 next year → they will not fund the project.

Risk Shifting vs. Debt Overhang

Two different agency costs of debt.

RISK SHIFTING — OVERINVESTMENT

Shareholders may choose excessively risky projects.

Why? Upside goes to equity; downside is partly shifted to creditors.

Typical symptom: higher volatility, lower total NPV.

DEBT OVERHANG — UNDERINVESTMENT

Shareholders may reject positive-NPV projects.

Why? New investment mainly increases creditor recovery.

Typical symptom: valuable projects are not funded.

Both problems become more important when leverage is high.

Group Case — Helix Pharma: Take the Safe Project or Swing for the Upside?

A leveraged company must choose one of two mutually exclusive R&D projects.

CURRENT POSITION

Debt repayment in one year: €400m

Existing assets in one year: €250m

The firm can invest €100m today in ONE project.

Shareholders choose the project.

PROJECT S — SAFE

80% chance: €300m

20% chance: €150m

Expected project payoff:
€270m

Strong probability of producing useful commercial value.

PROJECT R — RISKY

30% chance: €700m

70% chance: €0m

Expected project payoff:
€210m

A breakthrough outcome would be extremely valuable.

Which project maximizes firm value? Which might shareholders prefer? What conflict does this reveal?

Debt Can Also Create Agency Benefits

Leverage can discipline managers and reduce wasteful use of free cash flow.

LESS FREE CASH

Interest and principal commitments reduce cash managers can spend on weak projects.

LESS EMPIRE BUILDING

Debt can constrain acquisitions or expansion undertaken mainly to increase managerial control.

MORE OWNER DISCIPLINE

Using debt rather than new equity can allow owners to retain stronger incentives and control.

THREAT OF FAILURE

The possibility of losing control or employment can increase managerial effort.

Debt is therefore both a source of agency costs and a potential governance mechanism.

Worked Example — Rest-&Shirk Co.

Can debt induce a lazy owner-manager to work harder?

CURRENT FIRM

Current value = \$1,000

New project needs \$3,000

Interest rate = 10%

PROJECT CASH FLOW

High effort → \$5,000

Low effort → \$4,000

Low effort gives manager a private benefit of \$500.

INCENTIVE QUESTION

If financed with equity, will the manager work hard?

If financed with debt, does the answer change?

Think about the manager's total payoff: financial wealth + private benefit.

Rest-&-Shirk — Equity Financing

With outside equity, the manager prefers low effort.

HIGH EFFORT

Total firm value = \$6,000

Outside investors contribute \$3,000

Manager owns 25%

Manager's equity value = \$1,500

Private benefit = \$0

Total = \$1,500

LOW EFFORT

Total firm value = \$5,000

Manager owns 25%

Manager's equity value = \$1,250

Private benefit = \$500

Total = \$1,750

Manager chooses low effort: $\$1,750 > \$1,500$.

Rest-&-Shirk — Debt Financing

Debt can strengthen the manager's incentive to exert high effort.

HIGH EFFORT

Debt repayment = \$3,300

Equity value = \$2,700

Private benefit = \$0

Manager total payoff = \$2,700

LOW EFFORT

Debt repayment = \$3,300

Equity value = \$1,700

Private benefit = \$500

Manager total payoff = \$2,200

With debt, high effort becomes privately optimal: $\$2,700 > \$2,200$.

Generalized Value of a Levered Firm

A more complete capital-structure framework includes both agency costs and agency benefits.

GENERALIZED VALUE

$$VL = VU + PV(\text{Tax Shield}) - PV(\text{Distress Costs}) - PV(\text{Agency Costs}) + PV(\text{Agency Benefits})$$

TAX SHIELD

+ Benefit
Interest deductibility

DISTRESS

– Cost
Default and stakeholder
disruption

AGENCY COSTS

– Cost
Risk shifting and debt
overhang

AGENCY BENEFITS

+ Benefit
Managerial discipline

Optimal leverage balances all of these effects — not only taxes and bankruptcy.

Company Comparison — Who Should Use More Debt?

Apply the integrated framework, not a single ratio.

COMPANY X — SUPERMARKET

Stable demand
Tangible stores / inventory
Thin margins
Moderate growth
Predictable cash flow

COMPANY Y — BIOTECH

Negative current EBIT
R&D-intensive
Few tangible assets
High cash-flow uncertainty
Large growth options

DISCUSS

1. Which has more usable tax shield?
2. Which has higher distress cost?
3. Which has more debt capacity?
4. What evidence would you collect?

Asymmetric Information — Managers Know More Than Investors

Capital structure can also communicate information.

MANAGERS

Know more about customers, competitors, projects and future profitability.

INVESTORS

Observe public information but cannot perfectly verify management's private beliefs.

PROBLEM

A manager can say 'our stock is undervalued' — but every manager could say the same thing.

For a signal to be credible, it must be more costly for a weak firm to imitate.

Debt as a Signal

Why might a high-quality firm be more willing to issue debt?

GOOD FIRM

Expects strong future cash flows.
Can service additional debt with relatively low distress risk.

BAD FIRM

Imitating the same debt choice
creates a higher probability of default
and distress.

SIGNAL

Debt can be credible when imitation is sufficiently costly for the bad firm.

Important: debt is not always a reliable signal. The argument depends on the costs of imitation and managerial incentives.

Why Equity Issuance Can Be Interpreted Negatively

If managers believe shares are undervalued, they may avoid issuing equity.

MANAGER'S VIEW

Suppose managers believe the firm's true value is above the current market price.

EQUITY ISSUE

Selling new shares at the current price transfers value from existing shareholders to new investors.

INVESTOR INFERENCE

Investors may ask: 'Why are managers willing to sell shares now?' This can create an adverse-selection discount.

This creates a financing preference: internal funds → debt → equity.

Pecking Order Theory

Firms may follow a financing hierarchy rather than target a precise debt ratio.

1 INTERNAL FUNDS

Use retained earnings / cash first.

2 DEBT

If external financing is needed, issue relatively safe debt.

3 EQUITY

Issue equity last when information costs are greatest.

WHY?

Internal funds have no external adverse-selection problem.

Debt is less information-sensitive than equity.

Equity is most exposed to mispricing.

Group Case — LumiTech: Cash, Debt or Equity?

The project is attractive. The difficult question is how to finance it.

THE OPPORTUNITY

Investment required: €300m
Expected project NPV: +€90m

Current cash: €220m
Minimum operating cash buffer: €100m

The project should be undertaken.

FINANCING OPTIONS

- A. Use available cash
- B. Issue 5-year debt at 5.5%
- C. Issue new equity

Current debt is modest and the company has stable positive EBIT.

PRIVATE INFORMATION

Management believes shares are worth €50.
Current market price: €38.

Management also expects a major product announcement in 6 months but cannot disclose details yet.

Choose a financing plan. You may combine sources. Explain your choice using pecking order AND financial flexibility.

Trade-Off Theory vs. Pecking Order

Two theories can explain the same financing choice for different reasons.

TRADE-OFF THEORY

Core question: What debt level maximizes value?

Benefits: tax shield, managerial discipline

Costs: distress, agency costs

Prediction: firms move toward an optimal leverage range.

PECKING ORDER THEORY

Core question: Which financing source is least affected by information asymmetry?

Preference: internal funds → debt → equity

Prediction: leverage can reflect cumulative financing needs, not a fixed target.

In practice, both forces can operate at the same time.

In-Class Mini Case — Should NovaRetail Add Debt?

Work in groups for 10 minutes. Prepare a recommendation and two reasons.

COMPANY

Revenue €8.0bn
EBIT €640m
Stable demand
Tangible store network
Moderate growth
Tax rate 25%

CURRENT FINANCING

Debt €1.2bn
Cash €0.4bn
Interest €60m
Debt/EBIT $\approx 1.9\times$
Interest coverage $\approx 10.7\times$

PROPOSAL

Issue €800m debt to repurchase shares.

Discuss:

- Tax benefit?
- Distress risk?
- Agency effects?
- What extra data do you need?

Deliverable: Increase debt / Do not increase debt / Need more evidence — with justification.